Modeling, Typecasting, Formatting, and Renaming in Spark

In this train, we'll delve into modeling data, typecasting, formatting, and renaming columns in Spark.

Learning objectives

By the end of this train you should be able to:

- [Placeholder for objective 1]
- [Placeholder for objective 2]
- [Placeholder for objective 3]

Learning Objectives

- Model, typecast, format, and rename data in Spark.

Introduction

Our users will want to have the data prepared in a specific format, type, and structure. This last step in the data engineering toolkit is all about forming and manipulating the data into a structure and format that is acceptable for our end-users. Let's prepare our data for output onto a PowerBI dashboard. For this, we require the data in a wide format, with fields labeled with the full descriptions, and all field types as either string, float, or integer.

> **Definitions:** More on wide and narrow datasets and their uses can be found here.

```
working_df.printSchema()
```

Output:

```
root
 |-- county: string (nullable = true)
 |-- station: string (nullable = true)
 |-- latitude: float (nullable = true)
 |-- longitude: float (nullable = true)
 |-- date: timestamp (nullable = true)
 |-- rain: float (nullable = true)
 |-- temp: float (nullable = true)
 |-- wetb: float (nullable = true)
 |-- dewpt: float (nullable = true)
 |-- vappr: float (nullable = true)
 |-- rhum: integer (nullable = true)
 |-- msl: float (nullable = true)
 |-- wdsp: integer (nullable = true)
 |-- wddir: integer (nullable = true)
 |-- sun: float (nullable = true)
 |-- vis: integer (nullable = true)
 |-- clht: integer (nullable = true)
 |-- clamt: integer (nullable = true)
 |-- day: integer (nullable = true)
 |-- month: integer (nullable = true)
 |-- season: string (nullable = true)
 |-- idx: long (nullable = false)
```

Let's convert the date to a string. It will be parsed back to the timestamp on the dashboard.

```
working_df = working_df.withColumn('date', F.col('date').cast('STRING')) # We
can also use SQL notation in casting.
```

Let's convert the names to full names:

```
convert_dict = {
    'county': 'County',
    'station': 'Weather Station',
    'latitude': 'Latitude',
    'longitude': 'Longitude',
    'date': 'Date',
    'rain': 'Rain (mm)',
    'temp': 'Temperature (°C)',
    'wetb': 'Wet Bulb Air Temperature (°C)',
    'dewpt': 'Dew Point',
    'vappr': 'Vapour Pressure (hPa)',
    'rhum': 'Relative Humidity (%)',
    'msl': 'Mean Sea Level Pressure (hPa)',
    'wdsp': 'Mean Hourly Wind Speed (kt)',
    'wddir': 'Predominant Hourly Wind Direction (degrees)',
    'sun': 'Sun (hours)',
    'vis': 'Visibility (m)',
    'clht': 'Cloud Ceiling Height',
    'clamt': 'Cloud Amount (Oktas)',
    'idx': 'Index',
    'day': 'Day',
    'month': 'Month',
    'season': 'Season'}
```

```
output_df = working_df
for col in working_df.columns:
    output_df = output_df.withColumnRenamed(col, convert_dict[col])
```

```
output_df.printSchema()
```

Output:

```
root
 |-- County: string (nullable = true)
 |-- Weather Station: string (nullable = true)
 |-- Latitude: float (nullable = true)
 |-- Longitude: float (nullable = true)
 |-- Date: string (nullable = true)
 |-- Rain (mm): float (nullable = true)
 |-- Temperature (°C): float (nullable = true)
 |-- Wet Bulb Air Temperature (°C): float (nullable = true)
 |-- Dew Point: float (nullable = true)
 |-- Vapour Pressure (hPa): float (nullable = true)
 |-- Relative Humidity (%): integer (nullable = true)
 |-- Mean Sea Level Pressure (hPa): float (nullable = true)
 |-- Mean Hourly Wind Speed (kt): integer (nullable = true)
 |-- Predominant Hourly Wind Direction (degrees): integer (nullable = true)
 |-- Sun (hours): float (nullable = true)
 |-- Visability (m): integer (nullable = true)
 |-- Cloud Ceiling Height: integer (nullable = true)
 |-- Cloud Ammount (Oktas): integer (nullable = true)
 |-- Day: integer (nullable = true)
 |-- Month: integer (nullable = true)
 |-- Season: string (nullable = true)
 |-- Index: long (nullable = false)
```

That looks great!

Conclusion

In this lesson, you learned how to prepare and structure data in Spark by applying modeling techniques, typecasting columns, formatting values, and renaming fields for user-friendly outputs.

▼Key Takeaways ✖

- **Takeaway 1:** Data preparation is essential for usability in dashboards like PowerBI.
- **Takeaway 2:** Use `.cast()` to change data types such as converting dates to strings.
- **Takeaway 3:** Renaming columns can be done with `.withColumnRenamed()` in loops using dictionaries.
- **Takeaway 4:** Wide data format is often more suitable for visualization tools.
- **Takeaway 5:** Consistent formatting and labeling improves readability and usability of datasets.